
Exercise 1

1 Understanding LQR

Consider the following LQR formulation

$$\begin{aligned} \min_u \quad & \sum_{k=0}^{N-1} x_k^\top Q x_k + u_k^\top R u_k + x_N^\top P x_N, \\ \text{s.t.} \quad & x_{k+1} = A x_k + B u_k, \forall k \in [N-1]. \end{aligned}$$

1. Explain how the controller will behave when 1) $Q = 0, P = 0$. 2) $R = 0$.
2. Explain the difference between finite horizon LQR and infinite horizon LQR ($N = \infty$). Why is the infinite horizon controller time invariant?

2 Dynamic programming

Consider the following optimal control problem

$$\begin{aligned} V^*(x_0) = \min_u \quad & \gamma x_N^2 + \sum_{k=0}^{N-1} u_k^2, \\ \text{s.t.} \quad & x_{k+1} = x_k + u_k, \end{aligned}$$

where only the cost of the final state is penalized with $\gamma \geq 0$. Your goal is to solve this problem with dynamic programming.

1. Assume that $V_{k+1}(x_{k+1}) = h x_{k+1}^2$ and derive $V_k(x_k)$.
2. What is $V_N(x_N)$?
3. Compute the optimal value function $V^*(x_0)$ and the control law $\kappa^*(x_0)$ for a finite horizon N .
4. Compute the closed-loop pole of your system. Will your system converge to zero for all finite N and γ ?
5. What is your control law when $N \rightarrow \infty$. Will your system converge to zero for infinite N and finite γ ?